

$= \{x | x \in A \vee x \in B\}$ by definition of does not belong symbol
 $= \{x | x \in \bar{A} \vee x \in \bar{B}\}$ by definition of complement
 $= \{x | x \in A \cup B\}$ by definition of union
 $= A \cup B$ by meaning of set builder notation $\square$

Example 12: Prove the second distributive law from Table 1, which states that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ for all sets A, B and C.

Solution: We will prove this identity by showing that each side is a subset of the other side.

Let, $x \in A \cap (B \cup C)$. By Defn. 2, $x \in A \wedge x \in B \cup C$. By Defn. of AND, $x \in A$ and $x \in B \cup C$. By Defn. 1, $x \in B \vee x \in C$. By Defn. of OR, at least one of $x \in B$ and $x \in C$ is True.
 WLOG, Let $x \in B$. By Defn. of AND, $(x \in A \wedge x \in B)$. By Defn. 2, $x \in A \cap B$.
 By Defn. 1, $x \in A \cap B \cup (A \cap C)$.
 By Defn. of OR, $x \in A \cap B \vee x \in A \cap C$ is True.
 $\therefore$ By Defn. 3 of Sec-2.1, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Let, $x \in (A \cap B) \cup (A \cap C)$. By Defn. 1, $x \in (A \cap B) \vee x \in (A \cap C)$. By Defn. of OR, Atleast one of $x \in (A \cap B)$ or $x \in (A \cap C)$ is True.
 WLOG, Let $x \in (A \cap B)$ be True. By Defn. of AND, $x \in A \wedge x \in B$ is True.
 By Defn. 2, $x \in A \wedge x \in B$ is True. By Defn. of AND, $x \in A$ is True and $x \in B$ is True. By Defn. of OR, $x \in B \vee x \in C$ is True.
 By Defn. 1, $x \in (B \cup C)$ is True. By Defn. of AND, $x \in A \wedge x \in (B \cup C)$ is True.
 By Defn. 2, $x \in A \cap (B \cup C)$ is True.
 $\therefore$ By Defn. 3 of Sec-2.1, $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.
 $\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $\square$